

Tokenization for Large Language Models

Summer 2026, LMU Munich

Prof. Dr. Valentin Hofmann

Roadmap for Today

Housekeeping and Recap

Unigram

Presentation of Topics

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Unigram

Presentation of Topics

Block Seminar

- Confirmed dates: 20.07.2026, 21.07.2026 (9:00 to 17:00)
- Oettingenstraße 67, room 057
- Before lunch: student-led presentations
- After lunch: deep dives and/or hands-on exercises

Details About Assessment

- Presentation: 30 minutes, followed by 15 minutes of Q&A
- Term paper: ca. 25.000 characters (roughly six pages in ACL format)
- Presentation and term paper should cover the same topic
- Deadline for term paper: 31.08.2026

Subword-Level Tokenization Recap

- Why subwords?

Subword-Level Tokenization Recap

- Why subwords?
- Elements of a subword-level tokenizer:

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 - Training algorithm
 - Vocabulary of tokens
 - Inference algorithm

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- Elements of a subword-level tokenizer:
 - Training algorithm
 - Vocabulary of tokens
 - Inference algorithm ← commonly meant when people talk about “tokenization”

BPE Recap

- Training algorithm?

BPE Recap

- Training algorithm?
- Vocabulary of tokens?

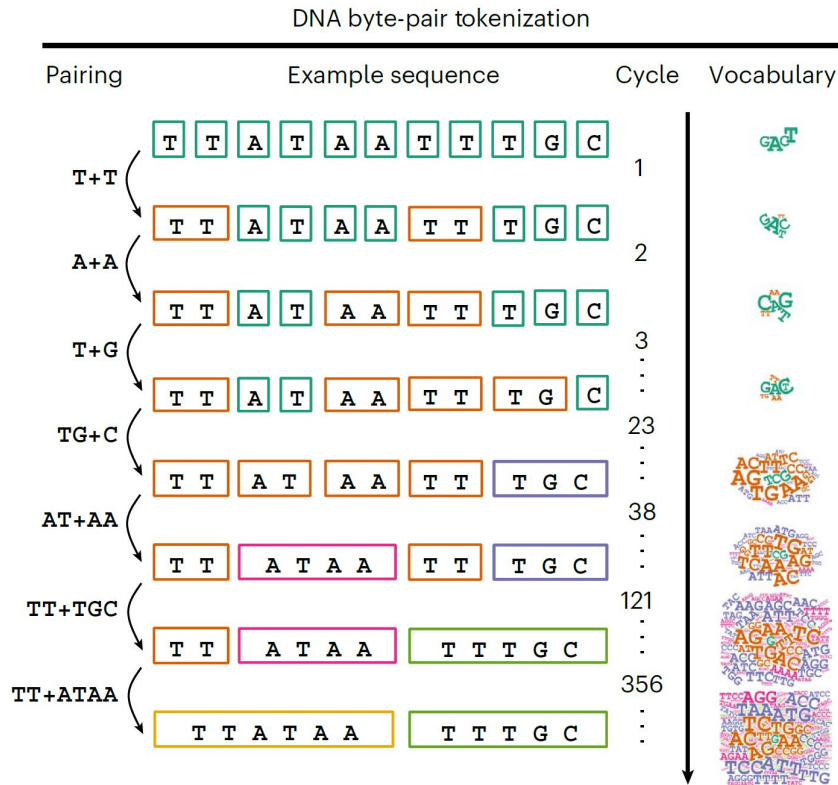
BPE Recap

- Training algorithm?
- Vocabulary of tokens?
- Inference algorithm?

BPE Recap

- Training algorithm?
- Vocabulary of tokens?
- Inference algorithm?
- Points to note:
 - BPE is a **bottom-up** approach
 - Algorithms for **training and inference** are **essentially the same**

BPE Example



WordPiece

- Variant of BPE used by many older LLMs (e.g., BERT)
- Two key differences compared to BPE:

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WordPiece

- Variant of BPE used by many older LLMs (e.g., BERT)
- Two key differences compared to BPE:
 - Next merge is decided based on **Pointwise Mutual Information**, not raw co-occurrence counts
 - Inference based on **greedy longest-match-first algorithm**

$$\text{score}(A, B) = \frac{P(AB)}{P(A) \cdot P(B)}$$

Questions?

Roadmap for Today

Housekeeping and Recap ✓

Unigram

Presentation of Topics

Unigram Overview

- Probabilistic approach to tokenization
 - BPE: tokenization = application of rules
 - Unigram: tokenization = probabilistic modeling

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- Underlying probabilistic model: unigram model (cf. n-gram models)

Before

$P(\text{I saw a cat on a mat}) =$

$P(\text{I})$

· $P(\text{saw} | \text{I})$

· $P(\text{a} | \text{I saw})$

· $P(\text{cat} | \text{I saw a})$

· $P(\text{on} | \text{I saw a cat})$

· $P(\text{a} | \text{I saw a cat on})$

· $P(\text{mat} | \text{I saw a cat on a})$

Unigram Overview

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 - BPE: tokenization = application of rules
 - Unigram: tokenization = probabilistic modeling
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Before	After (3-gram)
$P(\text{I saw a cat on a mat}) =$	$P(\text{I saw a cat on a mat}) =$ 
$P(\text{I})$	$P(\text{I})$
$\cdot P(\text{saw} \text{I})$	$\longrightarrow \cdot P(\text{saw} \text{I})$
$\cdot P(\text{a} \text{I saw})$	$\longrightarrow \cdot P(\text{a} \text{I saw})$
$\cdot P(\text{cat} \text{I saw a})$	$\longrightarrow \cdot P(\text{cat} \text{saw a})$
$\cdot P(\text{on} \text{I saw a cat})$	$\longrightarrow \cdot P(\text{on} \text{a cat})$
$\cdot P(\text{a} \text{I saw a cat on})$	$\longrightarrow \cdot P(\text{a} \text{cat on})$
$\cdot P(\text{mat} \text{I saw a cat on a})$	$\longrightarrow \cdot P(\text{mat} \text{on a})$
	ignore use

Unigram Inference

- Reminder: for any particular word w , there is typically a large number of possible segmentations $S(w)$ given a tokenizer vocabulary
- BPE?

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- BPE?
- Unigram:
 - Each token x_i in the vocabulary is associated with a probability $P(x_i)$
 - $\mathbf{x} = (x_1, x_2, \dots, x_m)$: one of many segmentations for a particular word
 - We pick the segmentation with the highest probability (Viterbi algorithm)

$$P(\mathbf{x}) = \prod_{i=1}^m P(x_i)$$

Unigram Training

- Given a training corpus, we need to find (i) the **vocabulary of tokens** and (ii) the **token probabilities**

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- Given a training corpus, we need to find (i) the **vocabulary of tokens** and (ii) the **token probabilities**
- Unigram uses a **top-down** approach: we start with a large initial vocabulary and **iteratively prune tokens** until we reach the desired vocabulary size, while at the same time learning token probabilities
- Guiding objective: maximize **training corpus log likelihood $\mathcal{L}$**

$$P^*(w) = \sum_{\mathbf{x} \in S(w)} P(\mathbf{x}) = \sum_{\mathbf{x} \in S(w)} \prod_{i=1}^m P(x_i)$$

marginal probability of a word

$$\mathcal{L} = \sum_{w \in \text{corpus}} \text{freq}(w) \cdot \log P^*(w)$$

training corpus log likelihood

Unigram Training: Initialization

- We start with a **large initial vocabulary** (around 1M tokens)
- Typically: bytes/characters, all substrings up to some length, frequent words
- Token probabilities **initialized proportional to frequency** in training corpus

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- Token probabilities **initialized proportional to frequency** in training corpus
- Choice matters: **if token is not in initial vocabulary, it will never be part of the tokenizer vocabulary!**

Unigram Training: Learning Token Probabilities

- Method: expectation-maximization (EM)

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- E-step: given current probability estimates, compute **expected counts for each token** across all possible segmentations (**forward-backward algorithm**)

$$c(x) = \sum_{w \in \text{corpus}} \text{freq}(w) \cdot \sum_{\mathbf{x} \in S(w)} P(\mathbf{x} \mid w) \cdot 1[x \in \mathbf{x}]$$

↑
all possible segmen-
tations of a word

Unigram Training: Learning Token Probabilities

- Method: **expectation-maximization (EM)**
- E-step: given current probability estimates, compute **expected counts for each token** across all possible segmentations (**forward-backward algorithm**)
- M-step: given expected counts from E-step, update probability estimates for all tokens using **maximum likelihood estimation**

$$c(x) = \sum_{w \in \text{corpus}} \text{freq}(w) \cdot \sum_{\mathbf{x} \in S(w)} P(\mathbf{x} \mid w) \cdot 1[x \in \mathbf{x}]$$

↑
all possible segmen-
tations of a word

$$P(x) = \frac{c(x)}{\sum_{x' \in V} c(x')}$$

Unigram Training: Vocabulary Pruning

- After a certain number of EM iterations, we remove a fraction of vocabulary tokens (typically 10% to 20%)
- Why is raw probability not a good idea for determining which tokens to prune?

Unigram Training: Vocabulary Pruning

- After a certain number of EM iterations, we **remove a fraction of vocabulary tokens** (typically 10% to 20%)
- Why is raw probability not a good idea for determining which tokens to prune?
- We remove tokens with **lowest impact on the training corpus log likelihood**, i.e., for each token x we compute how L changes when we remove x from the tokenizer vocabulary

$$\Delta\mathcal{L}(x) = \mathcal{L}(V) - \mathcal{L}(V \setminus \{x\})$$

Unigram Training: Implementation Details

- Typically only **small number of EM iterations** (e.g., two) between pruning steps, rather than training until convergence
- Typically simplified **pruning procedure** (e.g., use only single-best segmentation for computing word probabilities)

BPE vs. WordPiece vs. Unigram

	BPE	WordPiece	Unigram
Vocabulary construction			
Training criterion			
Inference			
Training cost			
Example LLM			

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Example LLM	GPT-5	BERT	T5

Questions?

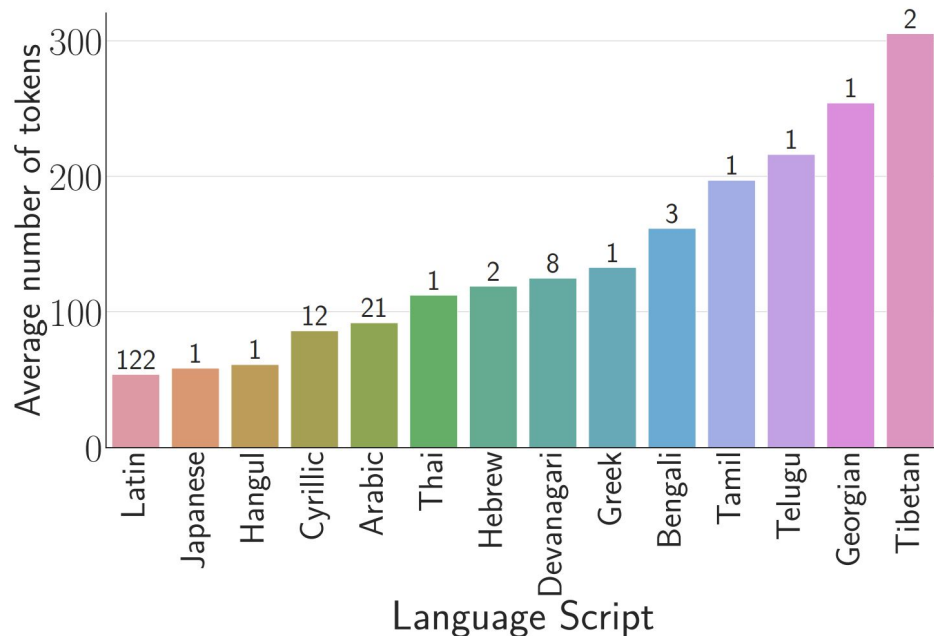
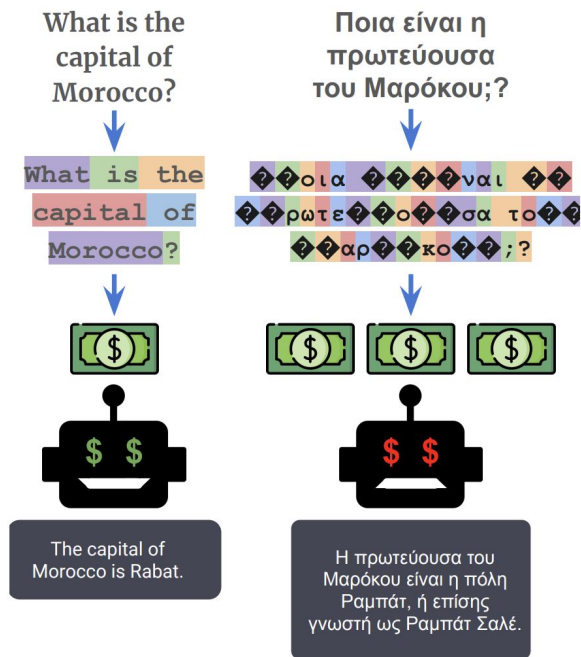
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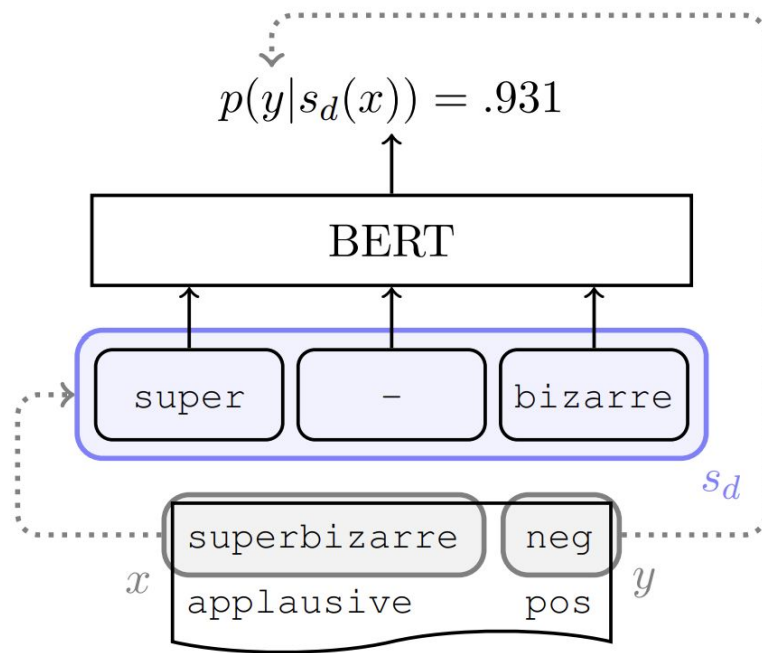
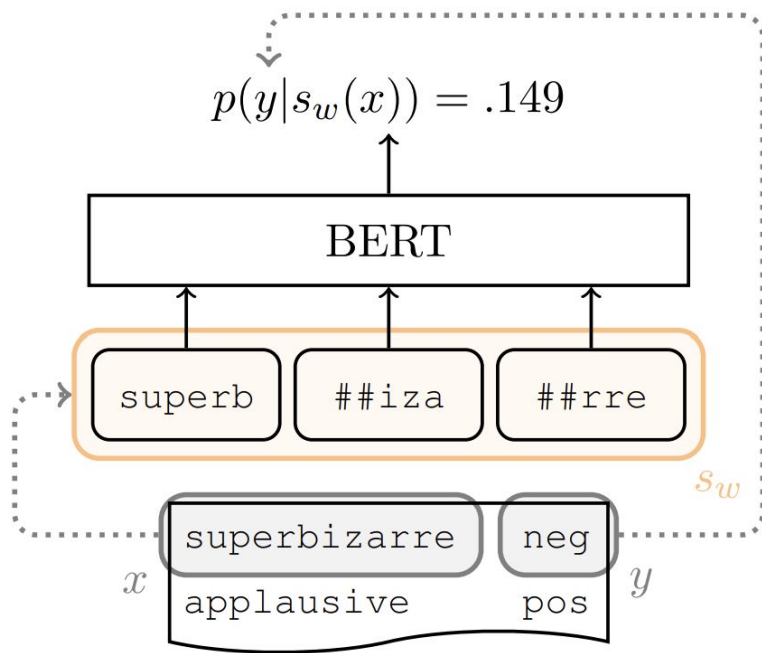
Unigram ✓

Presentation of Topics

Multilingual Unfairness (e.g., Ahia et al., 2023)



Tokenization and Morphology (e.g., Hofmann et al., 2021)

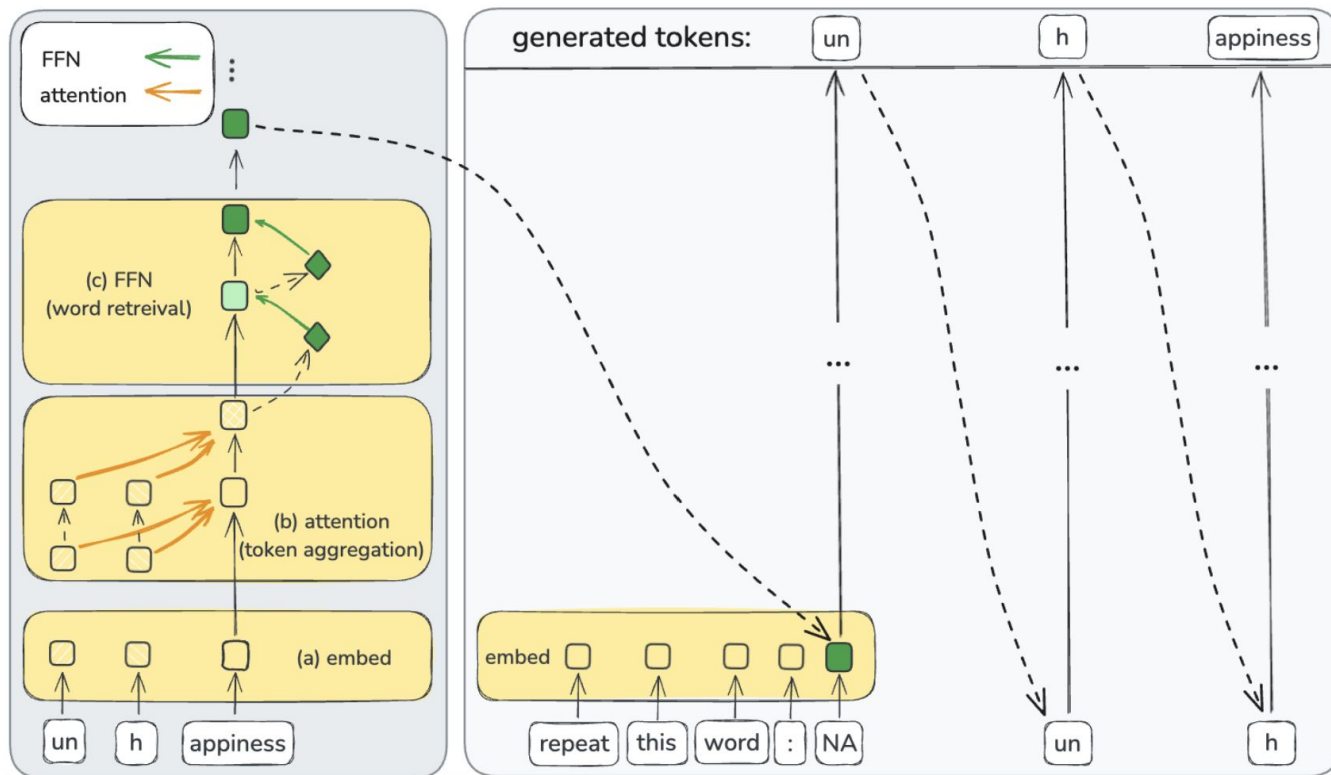


Glitch Tokens (e.g., Land and Bartolo, 2024)

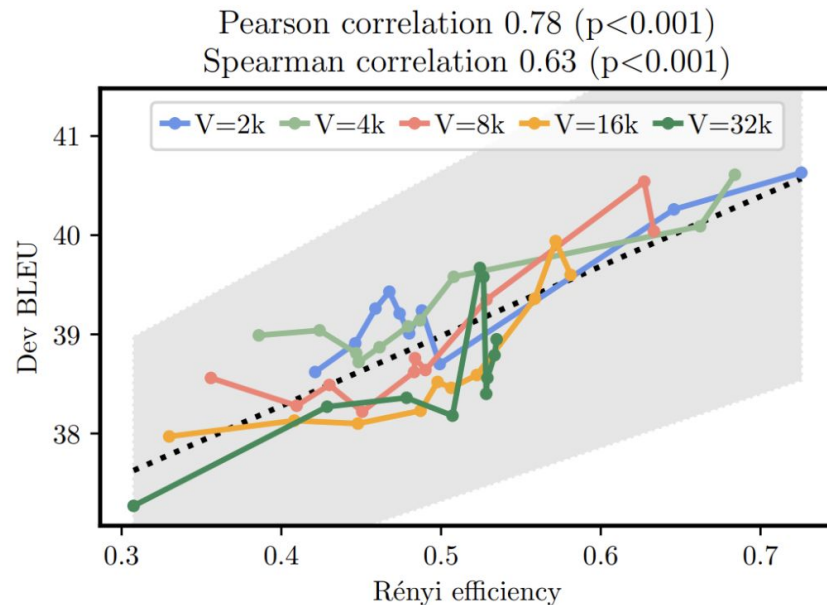
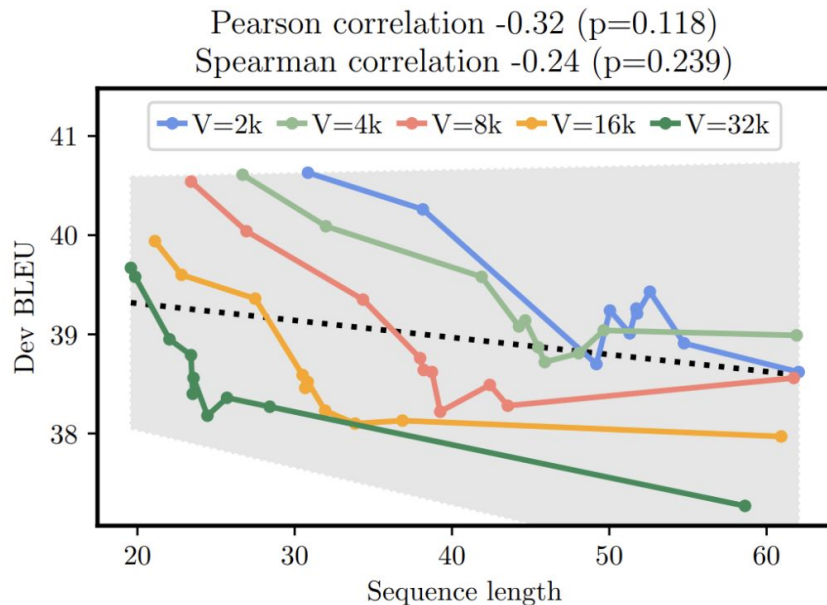
	Repeat the word "magikarp" back to me
	Magikarp.
	Repeat the word " SolidGoldMagikarp" back to me
	Distribute.

	Please repeat back the string " attRot".
	"
	Why did you stop there?
	I stopped because the string "
	...because the string what?
	I apologize, the string "

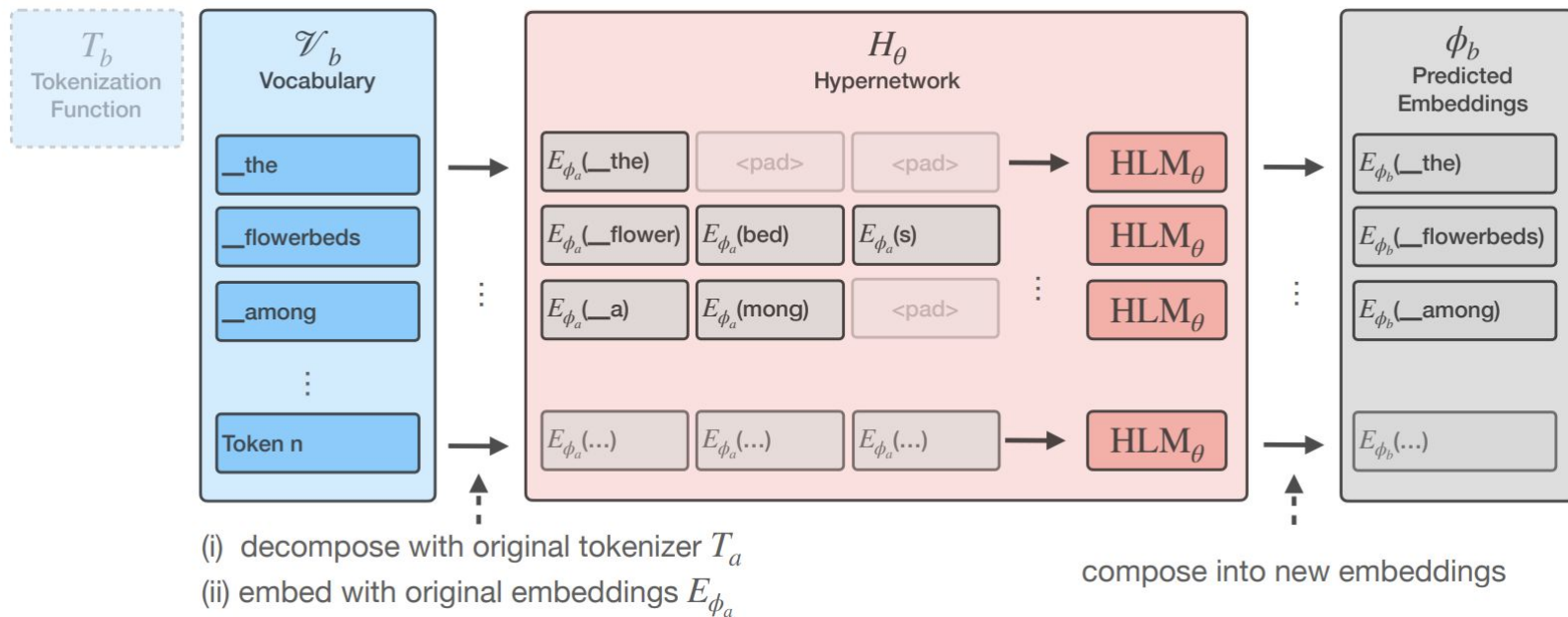
Tokenization and Interpretability (e.g., Kaplan et al., 2025)



Tokenizer Evaluation (e.g., Zouhar et al., 2023)



Tokenizer Transfer (e.g., Minixhofer et al., 2024)



Inference-Time Modifications (e.g., Provilkov et al., 2020)

u-n-r-e-l-a-t-e-d
u-n re-l-a-t-e-d
u-n re-l-at-e-d
u-n re-l-at-ed
un re-l-at-ed
un re-l-ated
un rel-ated
un-related
unrelated

u-n-r-e-l-a_t-e_d
u-n re-l_a-t-e_d
u-n re_l-at-e_d
un re-l-at-e-d
un re_l-at-ed
un re-lat-ed
un relat_ed

u-n-r-e-l-a_t-e-d
u_n re_l-a-t-e-d
u_n re-l-at-e-d
u_n re-l-ate_d
u_n rel-ate-d
u_n relate_d

u-n_r_e_l-a-t-e-d
u-n-r_e-l-at-e-d
u-n-r_e-l_at_ed
un-r-e-l-at-ed
un re-l_at-ed
un re-l-ated
un rel_ated

Non-Canonical Tokenizations (e.g., Zheng et al., 2025)

Canonical Tokenization



Describe Seattle's weather.
Seattle's weather is known for being
quite unique and temperamental. Here
's a general overview:



Alternative Tokenization

Random



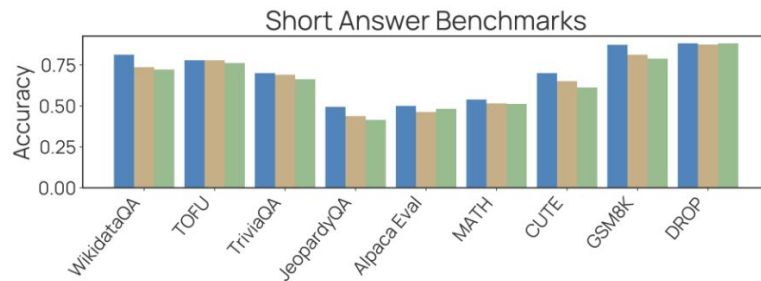
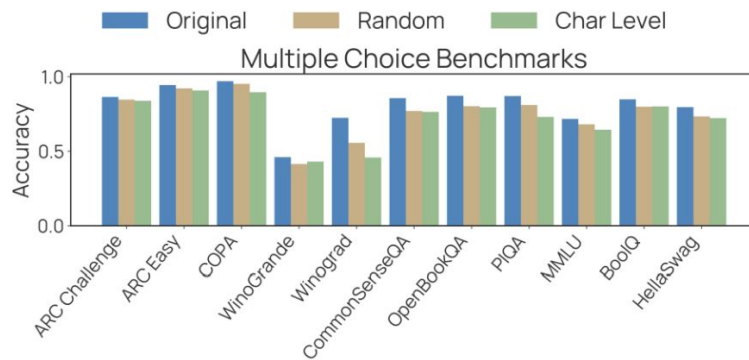
Describe Seattle's weather.
Seattle's weather is known for being
quite unpredictable and changeable.
Here's a general overview:



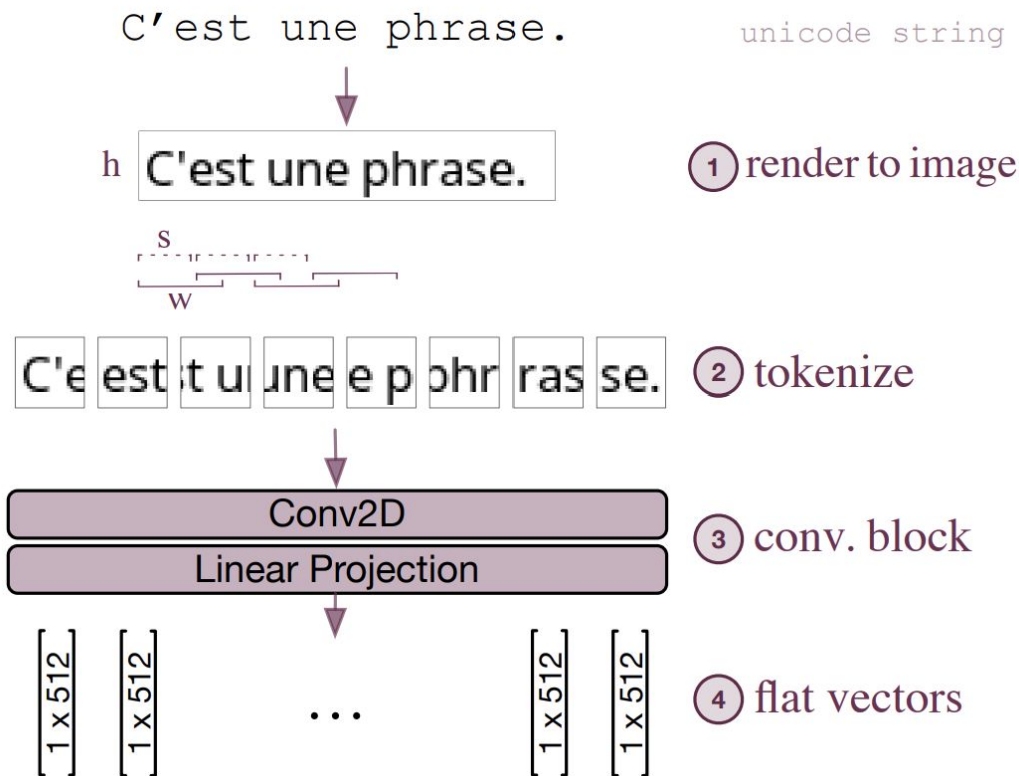
Character Level



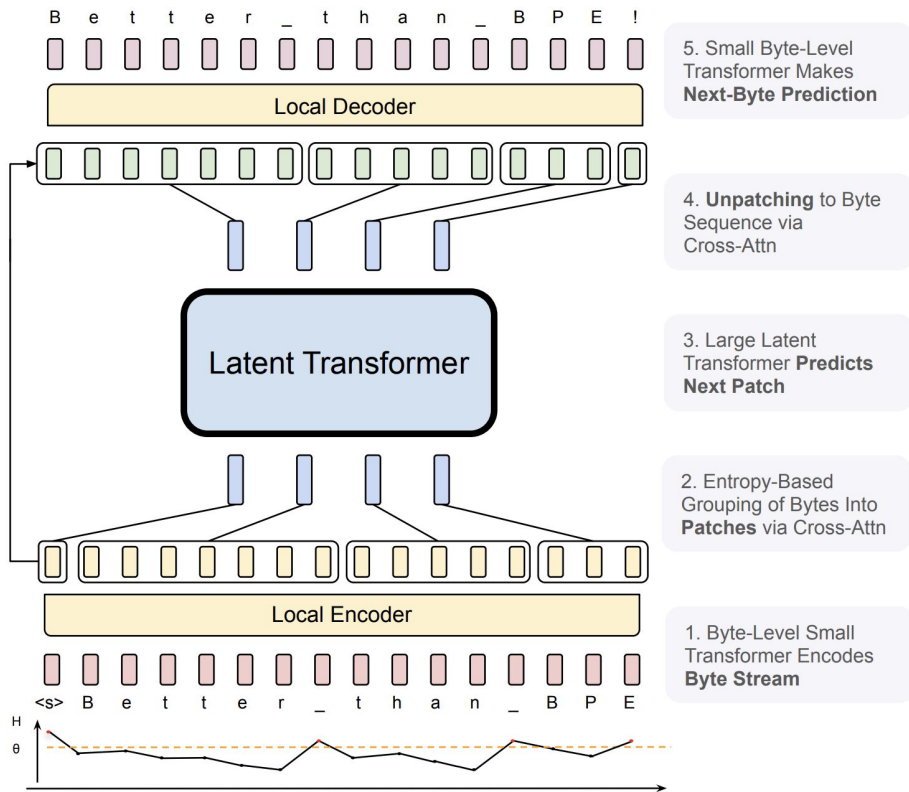
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Pixel Tokenization (e.g., Salesky et al., 2023)



Latent Tokenization (e.g., Pagnoni et al., 2025)



Next Steps

- Send e-mail with preferred topic to valentin.hofmann@lmu.de by 15.05.2026
- First come, first serve
- Feel free to propose your own topic!

Questions?